

TIC 229832383

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	6.126774 d	VSX	no match
Epoch (t0)	1628.8621 BTJD	ASAS-SN	no match
Duration	0.2440 d	Gaia eclipsing-binary table	no match
Depth	2.196 % (21,958 ppm)	Gaia variable classification	no match
SNR	97.6	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	4 (gate >= 3)	Verdict:	NOVEL
Robust depth ratio	1.05 (gate >= 0.50)		
Robust depth sigma	11.2 (gate >= 7.0)		
Odd/even z-score	0.03 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 12 light-curve dip in this sector: 11.8 sigma t0 refined by 0.0050 phase

Difference-image peak SNR 25.91 (gate >= 5.0) Compactness 1.00 (gate >= 0.50)

Dimming centroid offset 0.09 px (2 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 229833423

0.54 px away, dTmag +2.83 - farther from the dimming than the target

